

Null Space of a Matrix

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Null Space Example

Let

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 3 & 6 \\ 3 & 1 & 2 \end{bmatrix}$$

Find $\text{Nul}(A)$.

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Null Space is a Subspace.

Theorem

Let $A \in \mathbb{R}^{m \times n}$, then $\text{Nul}(A)$ is a subspace of $\mathbb{R}^n$.

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2. Let $\mathbf{u}, \mathbf{v} \in \text{Nul}(A)$, then see that

$$A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v}$$

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So, $\mathbf{u} + \mathbf{v} \in \text{Nul}(A)$ meaning it is closed under addition

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So, $c\mathbf{u} \in \text{Nul}(A)$ meaning it is closed under multiplication.



Column Space of a Matrix

Definition

Column Space: The **column space** of a matrix $A = [\mathbf{a}_1 \ \dots \ \mathbf{a}_n] \in \mathbb{R}^{m \times n}$ is denoted as $\text{Col}(A)$ and is the set of all linear combinations of columns of A .

$$\text{Col}(A) = \text{Span}(\mathbf{a}_1, \dots, \mathbf{a}_n) = \{\mathbf{b} \in \mathbb{R}^m \mid A\mathbf{x} = \mathbf{b} \text{ has a solution}\}$$

Finding a Basis of $\text{Col}(A)$ Example

Let

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 3 & 6 \\ 3 & 1 & 2 \end{bmatrix}$$

We want to know what an element of $\text{Col}(A)$ looks like, so we solve $A\mathbf{x} = \mathbf{b}$ and determine what $\mathbf{b}$ has to look like!

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & b_1 \\ 2 & 3 & 6 & b_2 \\ 3 & 1 & 2 & b_3 \end{array} \right]$$

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Finding a Basis of $\text{Col}(A)$ Example Continued

So, all the systems that we can solve have the form of

$$\text{Col}(A) = \left\{ \mathbf{b} \in \mathbb{R}^3 \mid \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ 7b_1 - 2b_2 \end{bmatrix} \right\}$$

Which can be written as

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Meaning

$$\begin{bmatrix} 1 \\ 0 \\ 7 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix}$$

forms a basis of $\text{Col}(A)$

An Easier Way to Compute a Basis of $\text{Col}(A)$

1. Reduce to RREF
2. The columns of A corresponding to pivot columns form a basis of $\text{Col}(A)$.

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So, the first two columns of A are a basis for $\text{Col}(A)$!

Showing These are Both Bases

We will show that

$$\text{Span} \left(\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} \right) = \text{Span} \left(\begin{bmatrix} 1 \\ 0 \\ 7 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix} \right)$$

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If we define $b_1 = c_1 + c_2$, $b_2 = 2c_1 + 3c_2$, and $b_3 = 3c_1 + c_2$, then

$$7b_1 - 2b_2 = 7(c_1 + c_2) - 2(2c_1 + 3c_2)$$

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$$7b_1 - 2b_2 = 7(c_1 + c_2) - 2(2c_1 + 3c_2) = 7c_1 + 7c_2 - 4c_1 - 6c_2 = 3c_1 + c_2 = b_3$$

This is exactly what we said the systems we can solve look like!

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3. Let $\mathbf{u} \in \text{Col}(A)$, $c \in \mathbb{R}$. Therefore, there is some $\mathbf{x} \in \mathbb{R}^n$ such that $A\mathbf{x} = \mathbf{u}$. See that

$$A(c\mathbf{x}) = cA\mathbf{x} = c\mathbf{u}.$$

So, $c\mathbf{u} \in \text{Col}(A)$!

Col(A) and Nul(A) Practice

For the following matrix $A \in \mathbb{R}^{4 \times 3}$ find a basis for Col(A) and Nul(A).

$$A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & 5 \\ 2 & 5 & 9 \\ 1 & 5 & 7 \end{bmatrix}$$

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What do we notice about the dimension of these spaces?

Relating to Linear Transformations

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear map, then we define:

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Kernel: The **kernel** of T is the set of all $\mathbf{x} \in \mathbb{R}^n$ such that $T(\mathbf{x}) = \mathbf{0}_m$.

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This is just the column space of the matrix associated with T !

Row Space of a Matrix

Let $A \in \mathbb{R}^{m \times n}$, then we can write it as

$$A = \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_m \end{bmatrix}$$

Where each $r_k^\top \in \mathbb{R}^n$ is a **row** of A .

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In addition, $\dim(\text{Col}(A)) = \dim(\text{Span}(\mathbf{a}_1, \mathbf{a}_2)) = 2$, so our definition is consistent!

Rank-Nullity Theorem

Theorem

Let $A \in \mathbb{R}^{m \times n}$, then we know that

$$\text{rank}(A) + \dim(\text{Nul}(A)) = n$$

$$\text{rank}(A) + \dim(\text{Nul}(A^\top)) = m$$